

VULNERABILITY ASSESSMENT REPORT

Future Interns – Cyber Security Internship Task 2 (2026)

Phishing Email Detection & Awareness

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1. INTRODUCTION

What is Phishing?

Phishing is a cyber attack technique where attackers send fake emails pretending to be trusted organizations such as banks, social media platforms, or payment services. The goal is to trick victims into revealing sensitive information such as usernames, passwords, banking details, or payment information.

Attackers commonly use fake login pages, malicious links, and urgent messages to manipulate users into taking immediate action without verifying the legitimacy of the email.

Why is Phishing Dangerous?

Phishing attacks are dangerous because:

- They can steal personal and financial information
- They may lead to identity theft
- They can compromise business systems
- One successful attack can result in financial loss
- Employees may unknowingly expose company credentials

Studies show that phishing remains one of the leading causes of cyber security breaches worldwide.

Purpose of This Report

- Analyze phishing email samples
- Identify common phishing indicators
- Explain how phishing attacks work
- Educate users on recognizing phishing attempts
- Provide phishing prevention guidelines

2. EXECUTIVE SUMMARY

Overview

In this task, three phishing email samples were analyzed:

1. Social Media Account Verification Scam
2. Payment Notification Scam
3. Bank Security Alert Scam

Each email was examined to identify phishing indicators, malicious behavior, and attack techniques used by cybercriminals.

Main Findings

All analyzed emails were confirmed to be phishing attacks because they contained:

- Fake sender domains
- Suspicious links
- Generic greetings
- Urgency and fear tactics
- Requests for sensitive information
- Suspicious formatting and grammar mistakes

Overall Risk Level

Email Type	Classification	Risk Level
Social Media Verification	PHISHING	HIGH RISK
Payment Notification	PHISHING	HIGH RISK
Bank Security Alert	PHISHING	HIGH RISK

3. COMMON PHISHING RED FLAGS

3.1 Fake Sender Domains

Attackers create fake email domains that resemble legitimate organizations.

Examples

- security@social-verification-alert.com
- billing@secure-paymentnotice.com
- alerts@bank-securityteam.com

Legitimate organizations usually use official domains that exactly match their company name.

3.2 Suspicious Links

Phishing emails often contain malicious links leading to fake login pages.

Examples

- social-account-check.xyz
- payment-review-secure.xyz
- secure-bank-login.xyz

Users should always inspect links carefully before clicking.

3.3 Generic Greetings

Most phishing emails use generic greetings because attackers do not know the victim's real name.

Examples

- “Dear User”
- “Dear Customer”
- “Dear Valued Client”

3.4 Urgency and Fear Tactics

Attackers attempt to create panic to force users into acting quickly.

Examples

- “Your account will be suspended”
- “Unauthorized payment detected”
- “Verify your account immediately”

3.5 Grammar and Spelling Errors

Many phishing emails contain grammatical mistakes and awkward wording.

Examples

- “Your account has been temporary suspended”
- “Confirm you are account now”

Professional companies rarely send poorly written emails.

4. DETAILED PHISHING EMAIL ANALYSIS

EMAIL 1: Social Media Account Verification Scam

Social Media Verification Email

From: security@social-mediaverify.net
Subject: Verify Your Social Media Account

Dear User,

We detected unusual login activity on your account. Please verify your account immediately to avoid suspension.

Verify your account here:
<http://socialmedia-secureverify.xyz/login>

Social Media Security Team

Email Description

This phishing email pretended to come from a social media company and warned the user that their account would be suspended unless verification was completed immediately.

Fake Sender

security@social-verification-alert.com

Suspicious Link

<http://social-account-check.xyz>

Main Objective

Steal user login credentials.

Red Flags Identified

- Fake sender domain
- Suspicious external link
- Generic greeting
- Threat of account suspension
- Urgent request for verification

How the Attack Works

Step 1

Victim receives a fake verification email.

Step 2

Victim clicks the malicious verification link.

Step 3

Victim is redirected to a fake login page.

Step 4

Victim enters username and password.

Step 5

Attackers steal account credentials.

Risk Classification

PHISHING – HIGH RISK

EMAIL 2: Payment Notification Scam

Payment Notification Email

From: billing@secure-paymentalerts.net
Subject: Payment Successfully Processed

Dear Customer,

Your payment of \$850 has been processed successfully. If you did not authorize this payment, please review it immediately.

View payment details:

<http://payment-review-secure.xyz/confirm>

Payment Support Team

Email Description

This phishing email claimed that a payment had been processed or that suspicious transaction activity had been detected.

Fake Sender

billing@secure-paymentnotice.com

Suspicious Link

<http://payment-review-secure.xyz>

Main Objective

Steal financial information or account credentials.

Red Flags Identified

- Fake billing domain
- Suspicious payment alert
- Fear-based messaging

- Malicious payment review link
- Generic greetings

How the Attack Works

Step 1

Victim receives payment notification email.

Step 2

Victim clicks payment review link.

Step 3

Fake website asks for card or login information.

Step 4

Victim enters sensitive information.

Step 5

Attackers steal financial credentials.

Risk Classification

PHISHING – HIGH RISK

EMAIL 3: Bank Security Alert Scam

Bank Security Alert Email

From: alert@secure-bankverify.net
Subject: Urgent Security Alert

Dear Customer,

We detected suspicious activity in your bank account. Please log in immediately to secure your account and avoid temporary suspension.

Secure your account here:

<http://bank-security-check.xyz/login>

Bank Security Department

Email Description

This phishing email impersonated a bank and warned users about suspicious activity detected on their account.

Fake Sender

alerts@bank-securityteam.com

Suspicious Link

<http://secure-bank-login.xyz>

Main Objective

Capture banking credentials.

Red Flags Identified

- Fake banking domain
- Urgent security warning
- Suspicious login page
- Request for sensitive information
- Fear-based language

How the Attack Works

Step 1

Victim receives security alert email.

Step 2

Victim clicks verification link.

Step 3

Fake banking login page appears.

Step 4

Victim enters login credentials.

Step 5

Attackers gain access to banking accounts.

Risk Classification

PHISHING – HIGH RISK

5. PREVENTION GUIDELINES FOR EMPLOYEES

Always Verify Sender Addresses

Carefully inspect the sender's email address before trusting any message.

Avoid Clicking Suspicious Links

Hover over links before clicking to inspect the destination URL.

Never Share Passwords Through Email

Legitimate organizations never ask users to provide passwords through email links.

Watch for Urgent Messages

Be cautious of emails demanding immediate action or threatening account suspension.

Use Multi-Factor Authentication (MFA)

Enable MFA whenever possible to improve account security.

Keep Software Updated

Ensure browsers, antivirus software, and operating systems remain updated.

6. WHAT TO DO IF YOU SUSPECT A PHISHING EMAIL

If you suspect an email is phishing:

1. Do not click any links
2. Do not download attachments
3. Report the email to the IT department
4. Delete the email immediately
5. Change passwords if you accidentally clicked a malicious link

7. CONCLUSION

This report analyzed three phishing email samples used by attackers to steal credentials and financial information.

All analyzed emails were identified as phishing attempts due to the presence of:

- Fake sender domains
- Suspicious links
- Generic greetings
- Fear and urgency tactics
- Requests for sensitive information

The analysis demonstrated how phishing attacks rely heavily on social engineering techniques to manipulate users into compromising sensitive information.

Employee awareness and security training remain essential in reducing phishing risks within organizations.

8. RECOMMENDATIONS

To reduce phishing risks, organizations should:

1. Conduct regular phishing awareness training
2. Educate employees on phishing indicators
3. Implement email filtering solutions
4. Enable multi-factor authentication
5. Encourage employees to report suspicious emails
6. Run phishing simulation exercises regularly